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Bidirectional Forwarding Detection (BFD)

Abstract

This document describes a protocol intended to detect faults in the bidirectional path between two forwarding engines, including interfaces, data link(s), and to the extent possible the forwarding engines themselves, with potentially very low latency. It operates independently of media, data protocols, and routing protocols.

Status of This Memo

This is an Internet Standards Track document.

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Information about the current status of this document, any errata, and how to provide feedback on it may be obtained at http://www.rfc-editor.org/info/rfc5880.

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1. Introduction

An increasingly important feature of networking equipment is the rapid detection of communication failures between adjacent systems, in order to more quickly establish alternative paths. Detection can come fairly quickly in certain circumstances when data link hardware comes into play (such as Synchronous Optical Network (SONET) alarms). However, there are media that do not provide this kind of signaling (such as Ethernet), and some media may not detect certain kinds of failures in the path, for example, failing interfaces or forwarding engine components.

Networks use relatively slow "Hello" mechanisms, usually in routing protocols, to detect failures when there is no hardware signaling to help out. The time to detect failures ("Detection Times") available in the existing protocols are no better than a second, which is far too long for some applications and represents a great deal of lost data at gigabit rates. Furthermore, routing protocol Hellos are of no help when those routing protocols are not in use, and the semantics of detection are subtly different -- they detect a failure in the path between the two routing protocol engines.

The goal of Bidirectional Forwarding Detection (BFD) is to provide low-overhead, short-duration detection of failures in the path between adjacent forwarding engines, including the interfaces, data link(s), and, to the extent possible, the forwarding engines themselves.

An additional goal is to provide a single mechanism that can be used for liveness detection over any media, at any protocol layer, with a wide range of Detection Times and overhead, to avoid a proliferation of different methods.

This document specifies the details of the base protocol. The use of some mechanisms are application dependent and are specified in a separate series of application documents. These issues are so noted.

Note that many of the exact mechanisms are implementation dependent and will not affect interoperability, and are thus outside the scope of this specification. Those issues are so noted.

1.1. Conventions Used in This Document

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC 2119 [KEYWORDS].

2. Design

BFD is designed to detect failures in communication with a forwarding plane next hop. It is intended to be implemented in some component of the forwarding engine of a system, in cases where the forwarding and control engines are separated. This not only binds the protocol more to the forwarding plane, but decouples the protocol from the fate of the routing protocol engine, making it useful in concert with various "graceful restart" mechanisms for those protocols. BFD may also be implemented in the control engine, though doing so may preclude the detection of some kinds of failures.

BFD operates on top of any data protocol (network layer, link layer, tunnels, etc.) being forwarded between two systems. It is always run in a unicast, point-to-point mode. BFD packets are carried as the payload of whatever encapsulating protocol is appropriate for the medium and network. BFD may be running at multiple layers in a system. The context of the operation of any particular BFD session is bound to its encapsulation.

BFD can provide failure detection on any kind of path between systems, including direct physical links, virtual circuits, tunnels, MPLS Label Switched Paths (LSPs), multihop routed paths, and unidirectional links (so long as there is some return path, of course). Multiple BFD sessions can be established between the same pair of systems when multiple paths between them are present in at least one direction, even if a lesser number of paths are available in the other direction (multiple parallel unidirectional links or MPLS LSPs, for example).

The BFD state machine implements a three-way handshake, both when establishing a BFD session and when tearing it down for any reason, to ensure that both systems are aware of the state change.

BFD can be abstracted as a simple service. The service primitives provided by BFD are to create, destroy, and modify a session, given the destination address and other parameters. BFD in return provides a signal to its clients indicating when the BFD session goes up or down.

3. Protocol Overview

BFD is a simple Hello protocol that, in many respects, is similar to the detection components of well-known routing protocols. A pair of systems transmit BFD packets periodically over each path between the two systems, and if a system stops receiving BFD packets for long enough, some component in that particular bidirectional path to the neighboring system is assumed to have failed. Under some conditions, systems may negotiate not to send periodic BFD packets in order to reduce overhead.

A path is only declared to be operational when two-way communication has been established between systems, though this does not preclude the use of unidirectional links.

A separate BFD session is created for each communications path and data protocol in use between two systems.

Each system estimates how quickly it can send and receive BFD packets in order to come to an agreement with its neighbor about how rapidly detection of failure will take place. These estimates can be modified in real time in order to adapt to unusual situations. This design also allows for fast systems on a shared medium with a slow system to be able to more rapidly detect failures between the fast systems while allowing the slow system to participate to the best of its ability.

3.1. Addressing and Session Establishment

A BFD session is established based on the needs of the application that will be making use of it. It is up to the application to determine the need for BFD, and the addresses to use -- there is no discovery mechanism in BFD. For example, an OSPF [OSPF] implementation may request a BFD session to be established to a neighbor discovered using the OSPF Hello protocol.

3.2. Operating Modes

BFD has two operating modes that may be selected, as well as an additional function that can be used in combination with the two modes.

The primary mode is known as Asynchronous mode. In this mode, the systems periodically send BFD Control packets to one another, and if a number of those packets in a row are not received by the other system, the session is declared to be down.

The second mode is known as Demand mode. In this mode, it is assumed that a system has an independent way of verifying that it has connectivity to the other system. Once a BFD session is established, such a system may ask the other system to stop sending BFD Control packets, except when the system feels the need to verify connectivity explicitly, in which case a short sequence of BFD Control packets is exchanged, and then the far system quiesces. Demand mode may operate independently in each direction, or simultaneously.

An adjunct to both modes is the Echo function. When the Echo function is active, a stream of BFD Echo packets is transmitted in such a way as to have the other system loop them back through its forwarding path. If a number of packets of the echoed data stream are not received, the session is declared to be down. The Echo function may be used with either Asynchronous or Demand mode. Since the Echo function is handling the task of detection, the rate of periodic transmission of Control packets may be reduced (in the case of Asynchronous mode) or eliminated completely (in the case of Demand mode).

Pure Asynchronous mode is advantageous in that it requires half as many packets to achieve a particular Detection Time as does the Echo function. It is also used when the Echo function cannot be supported for some reason.

The Echo function has the advantage of truly testing only the forwarding path on the remote system. This may reduce round-trip jitter and thus allow more aggressive Detection Times, as well as potentially detecting some classes of failure that might not otherwise be detected.

The Echo function may be enabled individually in each direction. It is enabled in a particular direction only when the system that loops the Echo packets back signals that it will allow it, and when the system that sends the Echo packets decides it wishes to.

Demand mode is useful in situations where the overhead of a periodic protocol might prove onerous, such as a system with a very large number of BFD sessions. It is also useful when the Echo function is being used symmetrically. Demand mode has the disadvantage that Detection Times are essentially driven by the heuristics of the system implementation and are not known to the BFD protocol. Demand mode may not be used when the path round-trip time is greater than the desired Detection Time, or the protocol will fail to work properly. See section 6.6 for more details.

4. BFD Control Packet Format

4.1. Generic BFD Control Packet Format

BFD Control packets are sent in an encapsulation appropriate to the environment. The specific encapsulation is outside of the scope of this specification. See the appropriate application document for encapsulation details.

The BFD Control packet has a Mandatory Section and an optional Authentication Section. The format of the Authentication Section, if present, is dependent on the type of authentication in use.

The Mandatory Section of a BFD Control packet has the following format:

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|Vers |  Diag   |Sta|P|F|C|A|D|M|  Detect Mult  |    Length     |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                       My Discriminator                        |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                      Your Discriminator                       |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                    Desired Min TX Interval                    |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                   Required Min RX Interval                    |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                 Required Min Echo RX Interval                 |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

An optional Authentication Section MAY be present:

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|   Auth Type   |   Auth Len    |    Authentication Data...     |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Version (Vers)

The version number of the protocol. This document defines protocol version 1.

Diagnostic (Diag)

A diagnostic code specifying the local system's reason for the last change in session state. Values are:

```
   0 -- No Diagnostic
   1 -- Control Detection Time Expired
   2 -- Echo Function Failed
   3 -- Neighbor Signaled Session Down
   4 -- Forwarding Plane Reset
   5 -- Path Down
   6 -- Concatenated Path Down
   7 -- Administratively Down
   8 -- Reverse Concatenated Path Down
9-31 -- Reserved for future use
```

This field allows remote systems to determine the reason that the previous session failed, for example.

State (Sta)

The current BFD session state as seen by the transmitting system. Values are:

```
0 -- AdminDown
1 -- Down
2 -- Init
3 -- Up
```

Poll (P)

If set, the transmitting system is requesting verification of connectivity, or of a parameter change, and is expecting a packet with the Final (F) bit in reply. If clear, the transmitting system is not requesting verification.

Final (F)

If set, the transmitting system is responding to a received BFD Control packet that had the Poll (P) bit set. If clear, the transmitting system is not responding to a Poll.

Control Plane Independent (C)

If set, the transmitting system’s BFD implementation does not share fate with its control plane (in other words, BFD is implemented in the forwarding plane and can continue to function through disruptions in the control plane). If clear, the transmitting system’s BFD implementation shares fate with its control plane.

The use of this bit is application dependent and is outside the scope of this specification. See specific application specifications for details.

Authentication Present (A)

If set, the Authentication Section is present and the session is to be authenticated (see section 6.7 for details).

Demand (D)

If set, Demand mode is active in the transmitting system (the system wishes to operate in Demand mode, knows that the session is Up in both directions, and is directing the remote system to cease the periodic transmission of BFD Control packets). If clear, Demand mode is not active in the transmitting system.

Multipoint (M)

This bit is reserved for future point-to-multipoint extensions to BFD. It MUST be zero on both transmit and receipt.

Detect Mult

Detection time multiplier. The negotiated transmit interval, multiplied by this value, provides the Detection Time for the receiving system in Asynchronous mode.

Length

Length of the BFD Control packet, in bytes.

My Discriminator

A unique, nonzero discriminator value generated by the transmitting system, used to demultiplex multiple BFD sessions between the same pair of systems.

Your Discriminator

The discriminator received from the corresponding remote system. This field reflects back the received value of My Discriminator, or is zero if that value is unknown.

Desired Min TX Interval

This is the minimum interval, in microseconds, that the local system would like to use when transmitting BFD Control packets, less any jitter applied (see section 6.8.2). The value zero is reserved.

Required Min RX Interval

This is the minimum interval, in microseconds, between received BFD Control packets that this system is capable of supporting, less any jitter applied by the sender (see section 6.8.2). If this value is zero, the transmitting system does not want the remote system to send any periodic BFD Control packets.

Required Min Echo RX Interval

This is the minimum interval, in microseconds, between received BFD Echo packets that this system is capable of supporting, less any jitter applied by the sender (see section 6.8.9). If this value is zero, the transmitting system does not support the receipt of BFD Echo packets.

Auth Type

The authentication type in use, if the Authentication Present (A) bit is set.

```
    0 - Reserved
    1 - Simple Password
    2 - Keyed MD5
    3 - Meticulous Keyed MD5
    4 - Keyed SHA1
    5 - Meticulous Keyed SHA1
6-255 - Reserved for future use
```

Auth Len

The length, in bytes, of the authentication section, including the Auth Type and Auth Len fields.

4.2. Simple Password Authentication Section Format

If the Authentication Present (A) bit is set in the header, and the Authentication Type field contains 1 (Simple Password), the Authentication Section has the following format:

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|   Auth Type   |   Auth Len    |  Auth Key ID  |  Password...  |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                              ...                              |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Auth Type

The Authentication Type, which in this case is 1 (Simple Password).

Auth Len

The length of the Authentication Section, in bytes. For Simple Password authentication, the length is equal to the password length plus three.

Auth Key ID

The authentication key ID in use for this packet. This allows multiple keys to be active simultaneously.

Password

The simple password in use on this session. The password is a binary string, and MUST be from 1 to 16 bytes in length. The password MUST be encoded and configured according to section 6.7.2.

4.3. Keyed MD5 and Meticulous Keyed MD5 Authentication Section Format

The use of MD5-based authentication is strongly discouraged. However, it is documented here for compatibility with existing implementations.

If the Authentication Present (A) bit is set in the header, and the Authentication Type field contains 2 (Keyed MD5) or 3 (Meticulous Keyed MD5), the Authentication Section has the following format:

```
    0                   1                   2                   3
    0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |   Auth Type   |   Auth Len    |  Auth Key ID  |   Reserved    |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |                        Sequence Number                        |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |                      Auth Key/Digest...                       |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
   |                              ...                              |
   +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Auth Type

The Authentication Type, which in this case is 2 (Keyed MD5) or 3 (Meticulous Keyed MD5).

Auth Len

The length of the Authentication Section, in bytes. For Keyed MD5 and Meticulous Keyed MD5 authentication, the length is 24.

Auth Key ID

The authentication key ID in use for this packet. This allows multiple keys to be active simultaneously.

Reserved

This byte MUST be set to zero on transmit, and ignored on receipt.

Sequence Number

The sequence number for this packet. For Keyed MD5 Authentication, this value is incremented occasionally. For Meticulous Keyed MD5 Authentication, this value is incremented for each successive packet transmitted for a session. This provides protection against replay attacks.

Auth Key/Digest

This field carries the 16-byte MD5 digest for the packet. When the digest is calculated, the shared MD5 key is stored in this field, padded to 16 bytes with trailing zero bytes if needed. The shared key MUST be encoded and configured to section 6.7.3.

4.4. Keyed SHA1 and Meticulous Keyed SHA1 Authentication Section Format

If the Authentication Present (A) bit is set in the header, and the Authentication Type field contains 4 (Keyed SHA1) or 5 (Meticulous Keyed SHA1), the Authentication Section has the following format:

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|   Auth Type   |   Auth Len    |  Auth Key ID  |   Reserved    |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                        Sequence Number                        |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                       Auth Key/Hash...                        |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                              ...                              |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Auth Type

The Authentication Type, which in this case is 4 (Keyed SHA1) or 5 (Meticulous Keyed SHA1).

Auth Len

The length of the Authentication Section, in bytes. For Keyed SHA1 and Meticulous Keyed SHA1 authentication, the length is 28.

Auth Key ID

The authentication key ID in use for this packet. This allows multiple keys to be active simultaneously.

Reserved

This byte MUST be set to zero on transmit and ignored on receipt.

Sequence Number

The sequence number for this packet. For Keyed SHA1 Authentication, this value is incremented occasionally. For Meticulous Keyed SHA1 Authentication, this value is incremented for each successive packet transmitted for a session. This provides protection against replay attacks.

Auth Key/Hash

This field carries the 20-byte SHA1 hash for the packet. When the hash is calculated, the shared SHA1 key is stored in this field, padded to a length of 20 bytes with trailing zero bytes if needed. The shared key MUST be encoded and configured to section 6.7.4.

5. BFD Echo Packet Format

BFD Echo packets are sent in an encapsulation appropriate to the environment. See the appropriate application documents for the specifics of particular environments.

The payload of a BFD Echo packet is a local matter, since only the sending system ever processes the content. The only requirement is that sufficient information is included to demultiplex the received packet to the correct BFD session after it is looped back to the sender. The contents are otherwise outside the scope of this specification.

Some form of authentication SHOULD be included, since Echo packets may be spoofed.

6. Elements of Procedure

This section discusses the normative requirements of the protocol in order to achieve interoperability. It is important for implementors to enforce only the requirements specified in this section, as misguided pedantry has been proven by experience to affect interoperability adversely.

Remember that all references of the form "bfd.Xx" refer to internal state variables (defined in section 6.8.1), whereas all references to "the Xxx field" refer to fields in the protocol packets themselves (defined in section 4).

6.1. Overview

A system may take either an Active role or a Passive role in session initialization. A system taking the Active role MUST send BFD Control packets for a particular session, regardless of whether it has received any BFD packets for that session. A system taking the Passive role MUST NOT begin sending BFD packets for a particular session until it has received a BFD packet for that session, and thus has learned the remote system’s discriminator value. At least one system MUST take the Active role (possibly both). The role that a system takes is specific to the application of BFD, and is outside the scope of this specification.

A session begins with the periodic, slow transmission of BFD Control packets. When bidirectional communication is achieved, the BFD session becomes Up.

Once the BFD session is Up, a system can choose to start the Echo function if it desires and the other system signals that it will allow it. The rate of transmission of Control packets is typically kept low when the Echo function is active.

If the Echo function is not active, the transmission rate of Control packets may be increased to a level necessary to achieve the Detection Time requirements for the session.

Once the session is Up, a system may signal that it has entered Demand mode, and the transmission of BFD Control packets by the remote system ceases. Other means of implying connectivity are used to keep the session alive. If either system wishes to verify bidirectional connectivity, it can initiate a short exchange of BFD Control packets (a "Poll Sequence"; see section 6.5) to do so.

If Demand mode is not active, and no Control packets are received in the calculated Detection Time (see section 6.8.4), the session is declared Down. This is signaled to the remote end via the State (Sta) field in outgoing packets.

If sufficient Echo packets are lost, the session is declared Down in the same manner. See section 6.8.5.

If Demand mode is active and no appropriate Control packets are received in response to a Poll Sequence, the session is declared Down in the same manner. See section 6.6.

If the session goes Down, the transmission of Echo packets (if any) ceases, and the transmission of Control packets goes back to the slow rate.

Once a session has been declared Down, it cannot come back up until the remote end first signals that it is down (by leaving the Up state), thus implementing a three-way handshake.

A session MAY be kept administratively down by entering the AdminDown state and sending an explanatory diagnostic code in the Diagnostic field.